

Anshul Kumar

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🌐 Portfolio / Live Demos

PROFESSIONAL SUMMARY

Aspiring Software Engineer specializing in **Distributed Systems** and **Backend Development**. Proficient in **Java, C++, and System Design**, with a proven track record of architecting scalable applications including a Remote Code Engine and Custom Redis implementation.

EDUCATION

Vellore Institute of Technology, Bhopal

Sep. 2023 – June 2027

Bachelor of Technology in Computer Science

CGPA: 9.15 / 10.0

- **Relevant Coursework:** Operating Systems, Computer Networks, DBMS, Object Oriented Programming, Distributed Systems

TECHNICAL SKILLS

Languages : Java (Core & Advanced), C++ (STL), JavaScript (ES6+), SQL

Backend & Systems : Spring Boot 3, Docker, Microservices, REST APIs, WebSockets, Linux

Core Concepts : System Design, Distributed Systems, Multithreading, TCP/IP Networking

Frontend & Tools : React.js, Tailwind CSS, Git/GitHub, Postman

FEATURED PROJECTS

Distributed Code Engine | *Java, Spring Boot, Docker*

Nov. 2025

- Architected a scalable remote code execution engine capable of compiling and running C++, Java, and Python code.
- Engineered **Docker containerization** to sandbox user code, neutralizing system-level threats and ensuring isolation.
- Orchestrated process management via `Java's ProcessBuilder` to stream real-time execution logs to the client interface.

MiniRedis (Multi-threaded Key-Value Store) | *Java, TCP*

Oct. 2025

- Built a high-performance in-memory key-value store from scratch, replicating core Redis functionality.
- Designed a **custom binary TCP protocol** using `Java Sockets` to optimize client-server communication payload.
- Implemented fine-grained locking on `ConcurrentHashMap` to ensure thread safety under heavy concurrent read/write loads.

L7 Load Balancer | *Java, HTTP Server*

Sep. 2025

- Developed a Layer 7 Reverse Proxy to efficiently distribute incoming HTTP traffic across backend server clusters.
- Implemented **Round Robin** and **Least Connections** algorithms to dynamically optimize request routing.
- Leveraged lock-free **Atomic counters** to track active connections, maximizing throughput while minimizing latency.

SyncDraw (Collaborative Whiteboard) | *React.js, Socket.io*

Aug. 2025

- Created a real-time collaborative whiteboard enabling sub-100ms latency drawing for multiple concurrent users.
- Deployed a **Pub/Sub architecture** via `WebSockets` to broadcast vector coordinates instantly across clients.

ACHIEVEMENTS & CODING

- Achieved **Global Rank 5831** in Codeforces Global Round 31 (Div. 1 + Div. 2) [Dec. 2025].
- **Codeforces Max Rating: 1135** (Handle: *anshullakra8*); solved 150+ algorithmic problems.
- Specialized in Dynamic Programming, Graph Theory, and Tree Algorithms.
- Consistently maintained a **9.15/10.0 CGPA** throughout B.Tech curriculum; Scored **93%** in Class XII.